Visualising Emotion

Does an AI drawing carry the feeling in the words that made it?

A complete explanation of the project

MSc Artificial Intelligence dissertation · Shre Hari Prem Anandh
Supervisor: Dr Chaona Chen · University of Sheffield

Every figure in this document was verified against the analysis on 1 September 2026

The project in one paragraph. At the Festival of the Mind 2024, about a thousand members of the public typed a few words about how they were feeling, and DALL·E drew each entry as a black-and-white line drawing. This project builds and validates a way to measure whether the drawing carries the feeling that was in the words. It finds that alignment is real and well above chance, that it is much stronger when the drawing contains a readable face than when it is a scene, that the generator is inconsistent with itself, and that the method used to measure it matters a great deal.

1 The question

You can already check whether an AI drew what a prompt asked for, provided the prompt names objects. Ask for *a red car* and a metric can look for a car and check that it is red. That is what most text-to-image evaluation does.

Feelings do not work that way. If somebody types *tired but happy*, there is no object to look for. There is no agreed answer key. And the feeling is mixed, which most emotion tools handle badly.

The precise question this project answers:

To what extent does an automatic image-emotion reader agree with prompt-derived emotion labels for AI-generated monochrome drawings, and how does that agreement vary by visual content and by repeated prompts?

That wording matters, and it is deliberately narrower than *does the drawing convey the feeling*. This project measures **agreement between two automatic systems** — one reading the words, one reading the drawing. It does not establish that a human viewer perceives the same feeling. That distinction is carried honestly through everything that follows.

2 Why the question is open

97 papers were collected, read and triaged; 73 were kept as core evidence and organised into twelve themes. The literature establishes four things that, taken together, leave a gap.

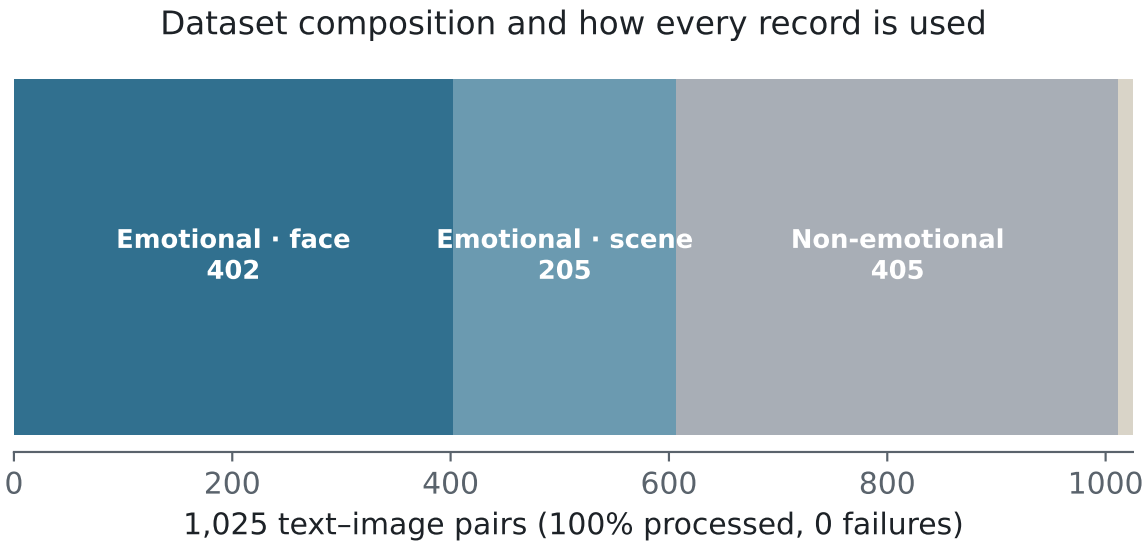
- **Emotion in text is well studied.** GoEmotions supplies a 27-category taxonomy, a large labelled corpus and an off-the-shelf classifier. But short, first-person self-report is the hardest register for these tools, and free self-report resolves roughly 27 blended categories rather than a tidy handful.
- **Emotional image *generation* is well studied.** Systems exist that deliberately make images carry a chosen emotion. Notably, the strongest evidence says lightness and chroma — not hue, and not shape — carry most of the emotional meaning in text-to-image output.

- **Text-to-image *evaluation* is well studied, but for objects.** The metrics score whether the picture contains what the prompt named. Recent work shows these metrics are fragile enough to be flipped by small perturbations.
- **Reading emotion back out of an image is possible but weak.** Vision-language models are positively biased and order-sensitive readers of emotion.

The gap. Nobody has checked whether the feeling in a person's own unedited words survives the journey into a drawing. This dataset is unusually well suited to asking: the prompts are real, first-person, short and often mixed; and because the drawings are monochrome line art, the channel the literature says usually carries emotion — colour — is simply absent.

An important restraint. The colour literature motivates why this dataset is a hard case. It is *not* a result of this project. There is no colour-control condition here, so no claim is made about the effect of removing colour. Every finding applies within this monochrome corpus.

3 The dataset, and what it took to use it



1,025 text–image pairs. Median prompt length is about four words. Before any analysis, three properties had to be handled.

- **Three incompatible filename schemes** in one folder: `img_00001.PNG` for ids 1–199, `img_200.PNG` for 200–999, `img_t1000.PNG` for 1000 and above — and two files use a lowercase extension. A naive loader resolves under 20% of the corpus; the loader written here resolves 100%.
- **Hidden provenance metadata.** Every PNG carries a C2PA/JUMBF manifest containing an RFC-3161 signing timestamp. Recovering it gave a creation time per image and revealed that **records 1–13 were generated before the festival** — developer tests. They are processed but excluded from analysis, so nothing is silently dropped.
- **Repeated prompts.** 87 distinct prompts were drawn more than once, covering 336 records. This looked like a nuisance and became a finding.

Dataset accounting	Records
Processed by the pipeline (0 failures)	1,025
Pre-festival developer tests, excluded	13
Analysed	**1,012**
of which the words carry a feeling	607
of which the words carry no feeling (the control)	405
drawings with a readable face	591
drawings with no readable face (scenes)	421
Distinct prompts (identity from pre-redaction text)	763
Records repeating an earlier prompt	249

4 The method

The supervisor's instruction was decisive: **never ask a vision model what emotion a picture shows.** Describe the picture in plain factual terms, then infer the feeling from that description with a separate model. The pipeline implements exactly that.

4.1 Stage by stage

1. **Load and redact.** Prompts are loaded through a single loader that applies name redaction, so the analysis, the figures and the demo app cannot diverge.
2. **Describe.** BLIP-VQA is asked five fixed questions about the drawing: *What is the facial expression?*, *What is the person doing?*, *What is the body posture?*, *How many people are there?*, *What is in the picture?* The answers are assembled into one neutral sentence. **The word "emotion" never appears in any question.**
3. **Extract, image side.** A zero-shot natural-language-inference model reads that description against the hypothesis template *This scene conveys* {}. across seven emotion families, returning a ranked distribution.
4. **Extract, text side.** GoEmotions reads the person's own words, and its 27 categories are collapsed onto the same seven families, so both sides live in one label space.
5. **Compare.** Four similarity measures, each referenced to chance.

4.2 Models used

Model	Size	Role
`Salesforce/blip-vqa-base`	∼385M	Describes the drawing (five questions)
`facebook/bart-large-mnli`	∼407M	Reads a feeling from that description
`SamLowe/roberta-base-go_emotions`	∼125M	Reads a feeling from the person's words
OpenCLIP ViT-B/32 (LAION-2B)	∼151M	*Frozen baseline only* — the older "ask CLIP directly" method, kept for comparison

4.3 The four similarity measures

Similarity, not matching — *happy* and *excited* should count as close, not as a miss.

1. **Valence–arousal similarity.** Each side's ranked emotions are collapsed to a single point on a pleasantness/energy plane using fixed centroid coordinates per emotion, and the measure is

one minus the normalised distance between the two points. *This is a designed proxy built on hand-set centroids, not a continuous lexicon rating.*
2. **Valence-sign agreement.** Do both sides land on the same side of pleasant/unpleasant?
3. **Emotion-family agreement.** Do both sides pick the same emotion family outright?
4. **Top-3 overlap.** How much of the two ranked distributions actually coincides?

5 What was tried and rejected

Each link was tested before being trusted. The failures shaped the design and are findings in their own right.

Approach	Why it was rejected (evidence)
GoEmotions on the image descriptions	Collapses to *neutral* above 0.95. It is trained on first-person feeling words; a third-person visual description contains none.
A single BLIP caption fed to the reader	Captions too terse (*a man with hands in the air*); the reader collapses to *surprise* for almost everything.
BART-MNLI on the *prompt* side	Over-reads *surprise* on short first-person text: 333 of 1,025 prompts came back as surprise. GoEmotions used instead.
Qwen2.5-VL-3B as the describer	7 GB in half precision exceeded a 16 GB laptop; thrashed into swap with **zero progress after 4.5 hours**. Never produced a description.
Florence-2 as the describer	Hard library incompatibility (`forced_bos_token_id`) on transformers 5.14.
Asking CLIP directly for the emotion	Beaten head-to-head (see Findings), and manual inspection found clear errors: arms-raised celebration read as *anger* 0.73.
BLIP-VQA (chosen)	$\sim$0.4B, roughly 1.5 s per record, and its "facial expression" answer supplies exactly the affective cue the next stage needs.

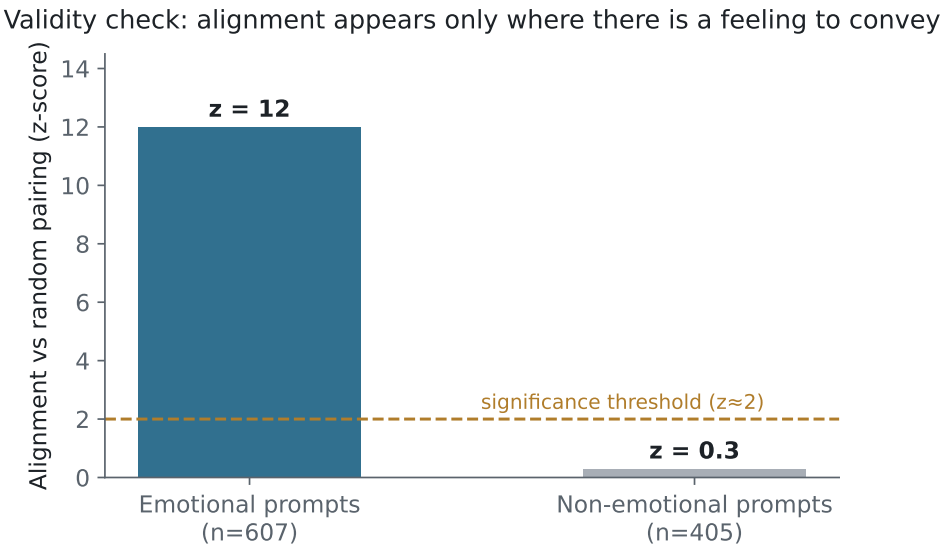
6 How the numbers are kept honest

This is the part that makes the findings usable, and it is where most of the methodological effort went.

6.1 A random-pairing null for every metric

Within each subset, the image side is shuffled and re-paired against the untouched prompt side, and all four metrics are recomputed. This is repeated **300 times** with a fixed seed. The average is the chance level; the spread gives a z-score. Every number reported in this project is quoted against its own null. A result counts only insofar as it beats chance.

6.2 A negative control that passes



405 of the 1,012 prompts contain no feeling at all — *pigs*, *mickey mouse*, *disney characters*. If the method were finding emotion that was not there, it would score above chance on these. It does not: valence–arousal similarity is 76.6% against a null of 76.6%, $z = +0.3$; emotion-family agreement sits exactly on its null.

> **Why this matters more than any single result.** A measurement instrument that reports a signal where there is none cannot be trusted anywhere. This control is the reason the other numbers can be believed.

6.3 A robustness check on repeated prompts

249 records repeat an earlier prompt, so treating all 1,012 as independent would overstate confidence. Every headline was re-run on the 763 distinct prompts only. All survive, at z between $+9$ and $+11$.

6.4 Degenerate nulls are refused

Where a subset shares one label by construction — for example the non-emotional records, which are all neutral — shuffling cannot change the score, so the "null" is an identity rather than a baseline. These are detected automatically and no chance figure is reported for them.

7 Chance is not zero

A late and important discovery, which changed how results are reported.

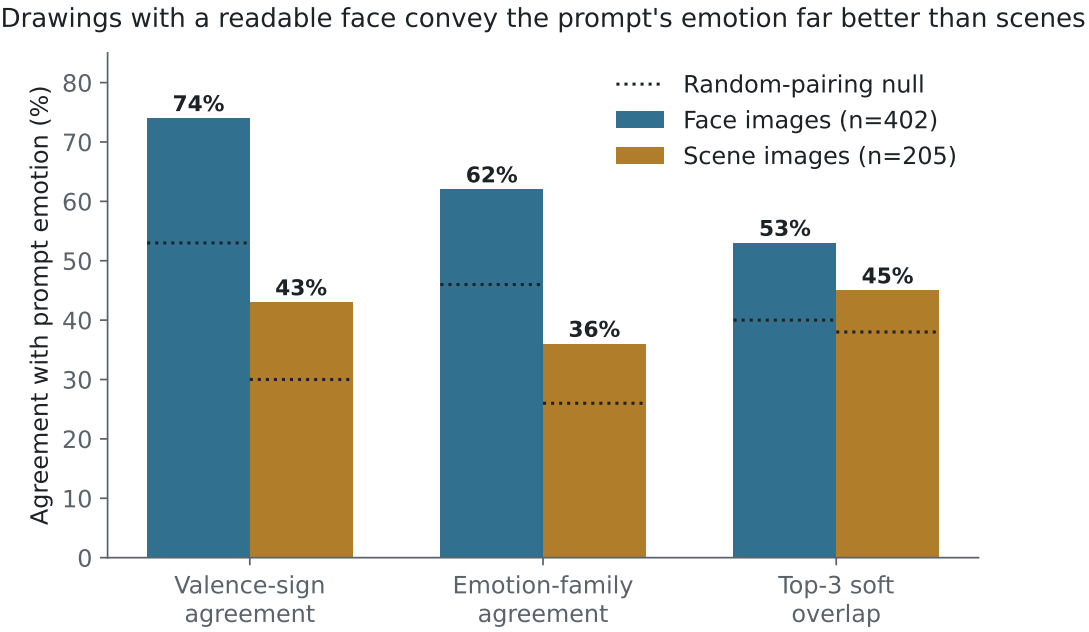
> The valence–arousal similarity measure **cannot fall below 41%** — that is its mathematical floor, given the fixed emotion coordinates. Two *randomly paired* records already score about **76–77%**, because most emotions sit close together on the plane.
>
> So a score of 71% looks like a pass and is in fact **below chance**. On a bare 0–100 scale that is invisible.

There is a second consequence. That measure returns roughly 84% for face drawings and 83% for scene drawings — it cannot tell them apart. The face/scene difference lives in valence-sign

and emotion-family agreement. The reporting was changed accordingly: those two measures lead, and the valence–arousal score is always shown against its measured chance line.

8 Findings

8.1 Alignment is real, and much stronger for drawings containing a face

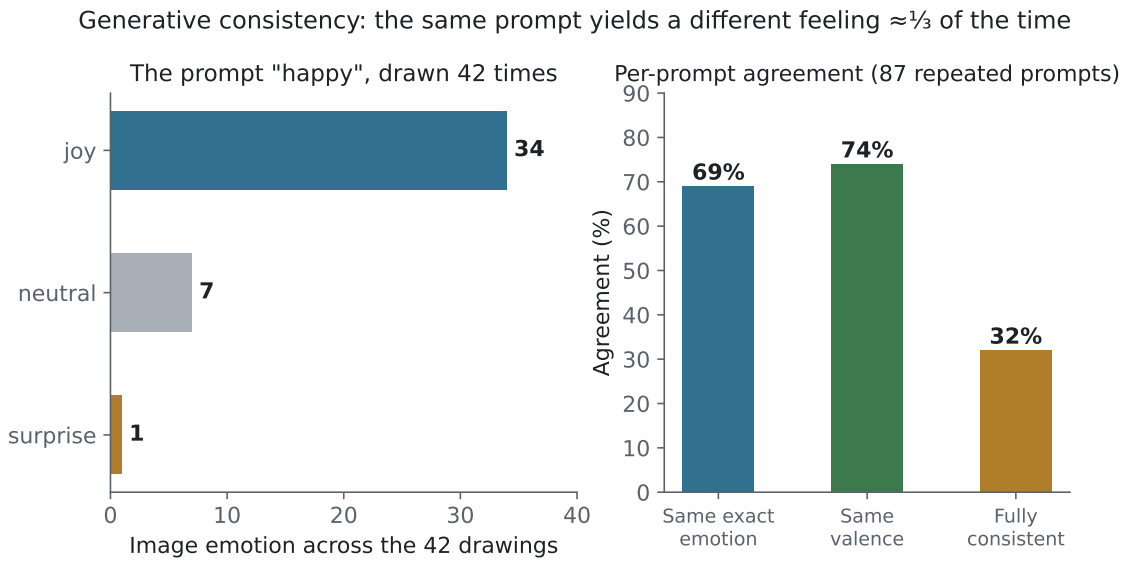


Emotional prompts only	Valence sign	Emotion family	Top-3 overlap
Face drawings ($n = 402$)	**73.9%** (z+10.4)	**61.9%** (z+10.3)	53.4% (z+11.6)
Scene drawings ($n = 205$)	42.9% (z+5.8)	35.6% (z+4.4)	44.9% (z+4.9)
All emotional ($n = 607$)	64.0% (z+11.1)	53.1% (z+10.3)	50.6% (z+11.6)

Both beat chance. Face drawings align about **1.7 times better** on both sign and family. On distinct prompts only, the gap holds: face 71.8% (z+8.5) against scene 37.3% (z+3.5).

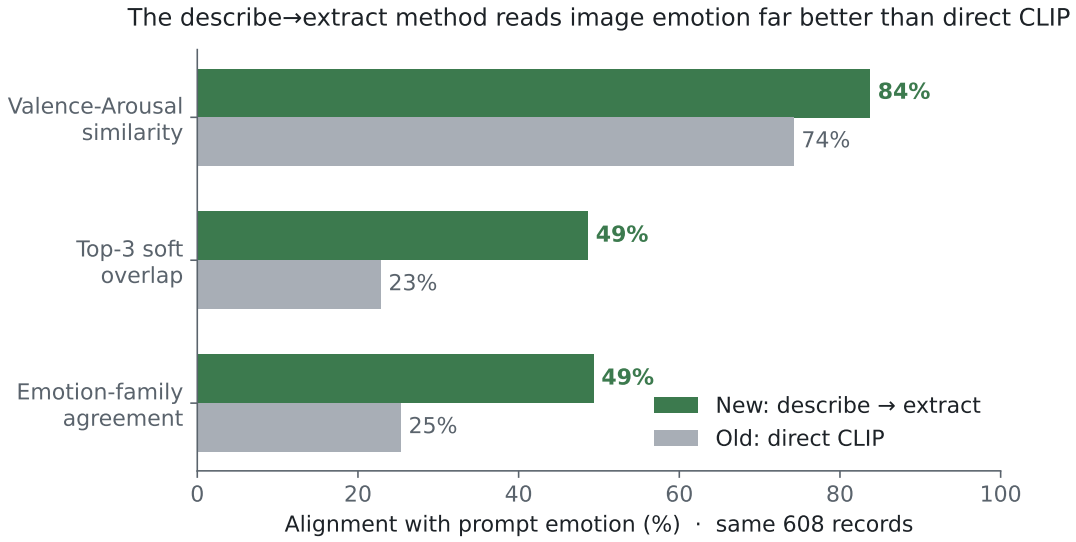
Stated carefully: the pipeline finds stronger prompt–image alignment for drawings it *classifies* as containing a readable face. It is not evidence that faces inherently carry emotion, because the classification and the emotion reading share a source — see Limitations.

8.2 The generator is inconsistent with itself



87 prompts were drawn between 2 and 42 times each, covering 336 records. Given *identical* words, the drawings convey the same emotion only **68.8%** of the time, and only **28 of the 87** repeated prompts are fully consistent across all their drawings. A single drawing is therefore one sample of the model's behaviour, not a fixed answer — which is itself a caution for anyone evaluating generative models on single outputs.

8.3 The method matters, and the supervisor's architecture wins



Head-to-head on the same records, with the prompt side held constant and only the image reader varied: describe-then-extract reaches **49.3%** emotion-family agreement against **25.3%** for asking CLIP directly, and 48.6% against 22.8% on top-3 overlap. The redirection was empirically justified rather than assumed.

9 The demo application

A small local web application runs the real pipeline live: choose a drawing from the dataset or upload an image, see the five factual answers, see both emotion readings side by side, and see

the similarity scored against chance. It exists to make the method inspectable rather than to collect anything.

Every number it shows is referenced to the measured chance line: the axis starts at the metric's real floor of 41, the middle half of random pairings is bracketed, and the headline reads *closer than N% of random pairings*. Nothing a visitor types or uploads is stored, and the request log is suppressed so no visitor IP addresses are recorded.

10 Ethics and data handling

No human participants were involved and no new data was collected, so no ethics approval was required. A human-rating study was designed in full and deliberately *not* run — approval alone would have consumed a fortnight of the remaining time. It stands as the clear next study.

10.1 Name redaction

Following the supervisor's direction — redact real names, keep illness and grief content because it is not identifying on its own and is essential to the analysis, and document the procedure — a three-stage pre-processing step was built.

1. **Detect**, with deliberately high recall. 77% of the corpus is written entirely in lower case, so capitalisation alone would miss most names. Candidates come from a 7,944-name gazetteer, from mid-sentence capitalisation checked against a ∼236,000-word dictionary, and from naming phrases such as *called X* or *my friend X*.
2. **Review.** Every candidate carries a recorded decision with a reason. Only names of private individuals are redacted; places, brands, public figures and medical eponyms such as Parkinson's and Alzheimer's are kept. The audit fails if any candidate is unreviewed.
3. **Apply**, scoped to the specific record each name was reviewed in.

Eight records contained names of private individuals; those become `[name]`. Two design lessons are recorded because both were real errors caught by the audit:

- **Unscoped redaction over-reaches.** A bare token list destroyed *Jack Frost* and *woolly mammoth*, because those strings occur innocently elsewhere.
- **Redaction can forge duplicates.** *karena cheung* and *Thomas chung* both become `[name]`, which made the analysis count two different people's entries as one prompt drawn twice. Duplicate identity is therefore decided on a key derived from the original text.

Redaction is analytically inert: re-running the text-emotion model on the redacted prompts moves each probability by at most 0.012 and changes **zero** top-1 emotion families.

Redaction is also a stronger position than withholding records. A hidden set is itself recoverable, because anyone can probe the public integer ids and read off which are missing. With names redacted, nothing needs hiding and all 1,012 records are shown.

11 Limitations — stated plainly

1. **This measures automated agreement, not human perception.** Both readers are models, and they disagree with each other in opposite directions: the CLIP baseline skews negative on line art, while the description model reports a happy or smiling expression for 57% of the faces it can read. Only human ratings could adjudicate. This is the single most important limitation.
2. **The face/scene split is partly circular.** It is defined by whether the description model reported an expression, and the emotion is read from that same description. Scene descriptions lack expression content, which mechanically lowers their scores. The effect survives the null and survives deduplication, but it is entangled with its own definition.

3. **The valence–arousal measure is a designed proxy**, built on hand-set emotion coordinates. It is not a continuous lexicon-based rating, and it cannot separate faces from scenes.
4. **No claim is made about colour.** The corpus is monochrome, but there is no colour condition to compare against.
5. **No positive-versus-negative direction is claimed.** An earlier claim was withdrawn: the single-valence subsets have degenerate nulls, and the two readers disagree about the direction.
6. **Statistical strengthening is still outstanding.** Confidence intervals, a paired test for the method comparison, and clustered resampling that treats repeated prompts as non-independent would all strengthen the results chapter.

12 Reproducibility

All analysis is driven by scripts with a fixed random seed and no manual steps. An automated test suite of **28 checks** covers the environment, data integrity, every analysis, a live run of the pipeline on fresh images, the redaction procedure and figure regeneration. It currently passes 28/28. One known weakness is recorded honestly: the figure script contains its numbers literally rather than reading them from result files, so it cannot detect a change in the analysis on its own.

13 What this contributes

The contribution is **not** a novel emotion classifier. It is: a question nobody had asked of real first-person self-report; a dataset that cannot be downloaded; an evaluation design with a working negative control and null baselines throughout; a demonstrated method comparison; a repeated-prompt consistency analysis that shows the generator is not deterministic in what it conveys; and a documented, tested data-protection procedure.

All figures and numbers in this document were regenerated and verified against the analysis on 1 September 2026. Authoritative results file: `outputs/RESULTS_final.md`.